

Neha K

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Objective

EEE undergraduate (CGPA 8.7) with hands-on experience in embedded systems and IoT hardware, seeking an internship in Semiconductor Test Engineering (STE). Eager to apply circuit theory, embedded C, and sensor validation skills in post-silicon validation and ATE-based test development.

Education

Siddaganga Institute of Technology	2023–2027
B.E. in Electrical & Electronics Engineering	CGPA: 8.7

Seshadripuram PU College	2023
Class XII (PUC)	94.1%

Chethana Vidya Mandira	2021
Class X (SSLC)	96.8%

Skills

Electronics: Circuit Theory, Analog/Digital Electronics, Network Analysis, Signal Fundamentals Embedded/IoT: Embedded C, Arduino, Raspberry Pi, Sensors, Actuators, Peripheral Interfacing
Programming: C++, Python (Basic) Web: HTML, CSS, JavaScript, React Databases: MySQL, Supabase Tools: Git, VS Code, TinkerCAD

Projects

- Automotive ECU Logic Simulator - Simulated ECU state-machine logic for automotive subsystems (engine, brake, lighting) using TinkerCAD and C-based firmware
- Real-Time Psychophysiological Stress Monitoring System - Interfaced GSR and pulse sensors with Raspberry Pi; implemented signal conditioning, real-time processing, and thresholdbased health alerts.
- LRU Cache Simulator - Implemented an LRU Cache in C++ using hash map and doubly linked list, achieving O(1) get/put operations.
- Customer Sentiment Alert System - Classified customer feedback sentiment using ML and triggered real-time alerts for negative responses, with Supabase (MYSQL) storage.
- Air Pollution Monitoring System - Tracked real-time air quality using MQ gas sensors and Arduino, with data logged to Supabase and threshold-based alerts.

Internship

Emertxe Information Technologies (2024)

- Trained in embedded C and microcontroller architecture with focus on peripheral interfacing and real-time control.
- Built IoT-based projects integrating sensors and actuators, from prototyping through testing.

Certificates

HTML5 Fundamentals (Ceeras), Cisco Python Essentials 1 & 2, CCNA: Cisco Certified Network Associate , Nutanix Certified Associate 6 (NUTANIX)

Achievements

- Finalist – Synaphack 3.0
- Participant – AI Agent Hackathon, IEEE Decodex

Leadership & Activities

Volunteer, IEEE Photonics and ComSoc – Board Member, IEEE Photonics and ComSoc – Spearheaded planning and end-to-end management of technical events and workshops, coordinating cross-functional teams to ensure successful execution.